

K-Means Clustering Walkthrough

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Imagine we have 4 data points on a 2D graph, and we want to group them into 2 clusters ($K = 2$).

Here are our four points:

- $P_1 = (1, 1)$
- $P_2 = (2, 1)$
- $P_3 = (4, 3)$
- $P_4 = (5, 4)$

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Step 1: Initialization

The algorithm must start somewhere. It takes our four raw data points and must randomly generate $K = 2$ initial cluster centers (centroids).

In this specific example, we told the algorithm to just use two existing points as the starting centroids: $P_1(1, 1)$ and $P_3(4, 3)$. The resulting initial setup is shown below.

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INITIAL STATE (STEP 1): STARTING POINTS AND INITIAL CENTROIDS



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Step 2: First Cluster Assignment

Now, the algorithm must "color" the points based on which initial star is closer. We calculate the distance from every point to both stars and assign the point to the closest one.

To make the calculation easy, we are using Squared Euclidean Distance: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.

- **P2 (2,1):** $(2 - 1)^2 + (1 - 1)^2 = \mathbf{1}$ (to Blue C_1). Distance to Red C_2 is 8. *P2 becomes Blue.*
- **P4 (5,4):** $(5 - 4)^2 + (4 - 3)^2 = \mathbf{2}$ (to Red C_2). Distance to Blue C_1 is 25. *P4 becomes Red.*

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FIRST ASSIGNMENT (STEP 2): COLORING CLUSTERS



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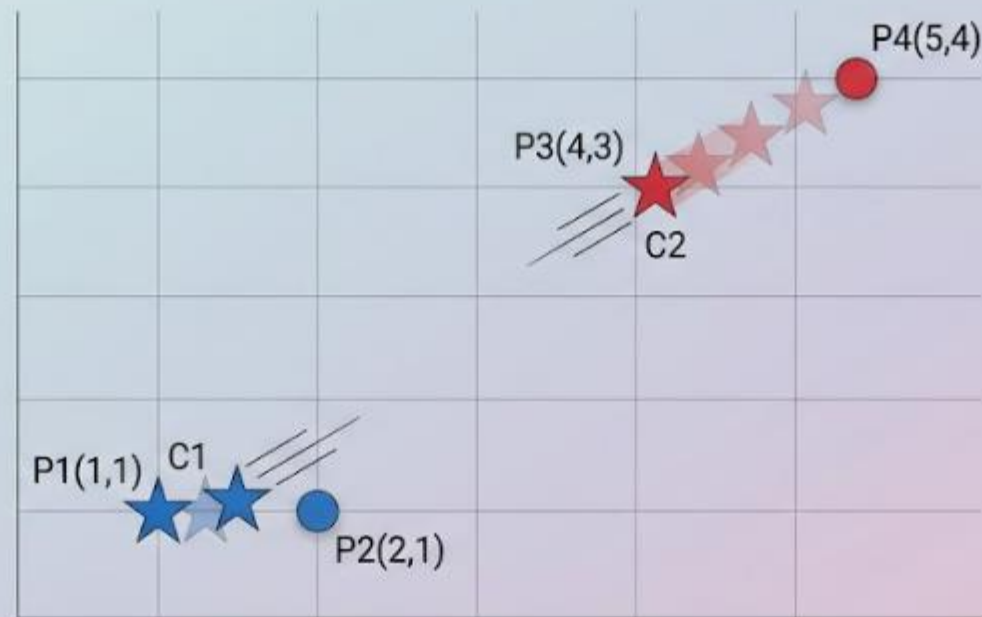
Step 3: Centroid Update (The Key Movement)

Because points have been assigned, the centroids (C_1 , C_2) are no longer in the "center" of their colored teams. The algorithm re-calculates the center of gravity for each group by finding the mean of their coordinates. This is the key iterative step.

- **New Blue Centroid (C_1):** Average of (1,1) and (2,1): $\left(\frac{1+2}{2}, \frac{1+1}{2}\right) = (1.5, 1)$
- **New Red Centroid (C_2):** Average of (4,3) and (5,4): $\left(\frac{4+5}{2}, \frac{3+4}{2}\right) = (4.5, 3.5)$

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CENTROID UPDATE (STEP 3): MOVING TO THE CENTER



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Steps 4 & 5: Convergence and Stability

Now that the centroids have settled into their new locations (1.5, 1 and 4.5, 3.5), we run the distance check *one more time*. We need to know if any points are now closer to the other "team's" star.

In our simplified example, because the gaps between the clusters are large, moving the centroids slightly closer to each other did not change which cluster is nearest to any data point.

- P1 is still closer to the new Blue C_1 .
- P3 is still closer to the new Red C_2 .

Because no data points decided to change teams, the algorithm has reached equilibrium. This is called **convergence**, and it now terminates. The clusters are stable.

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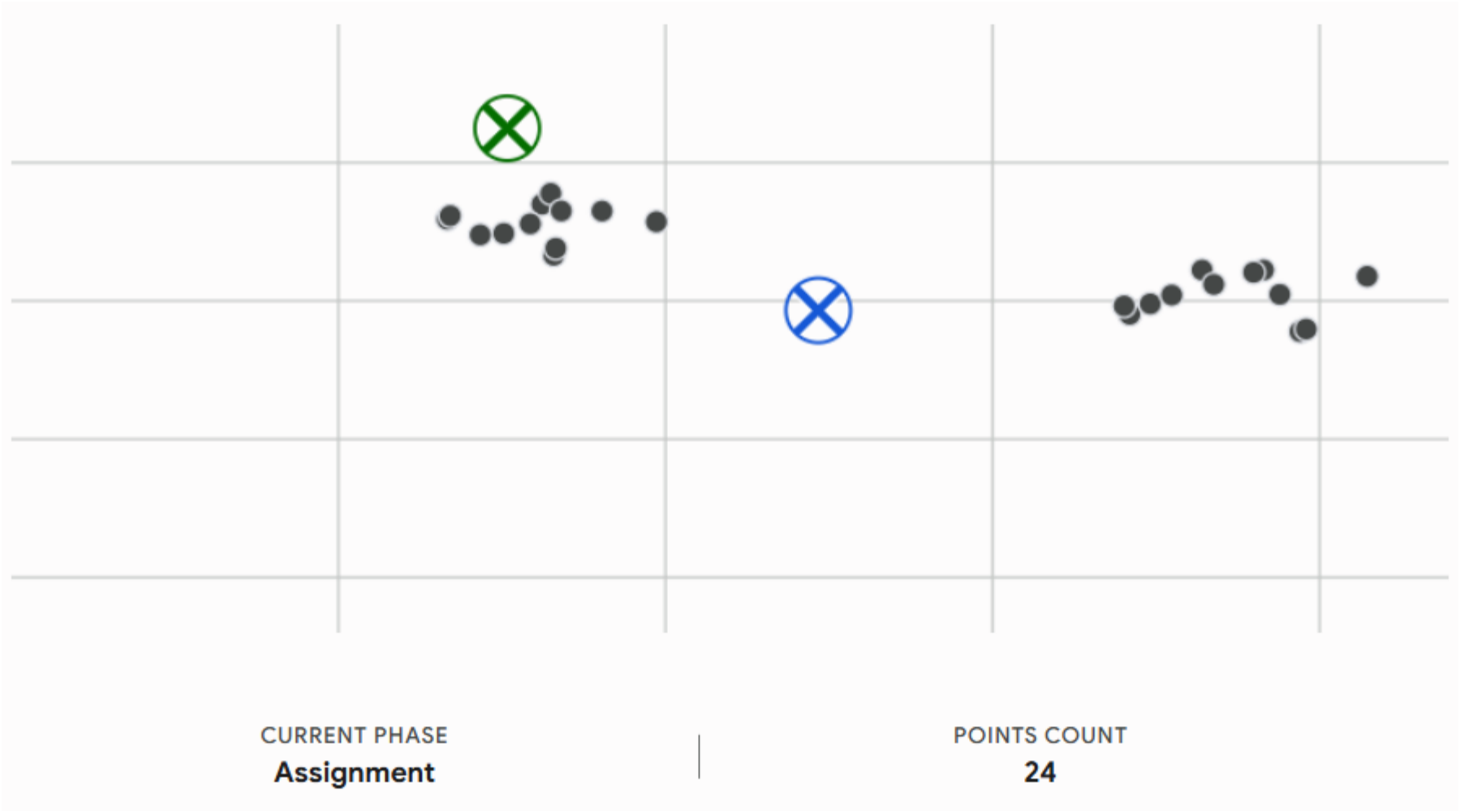
FINAL CONVERGENCE (STEPS 4 & 5): STABLE CLUSTERS



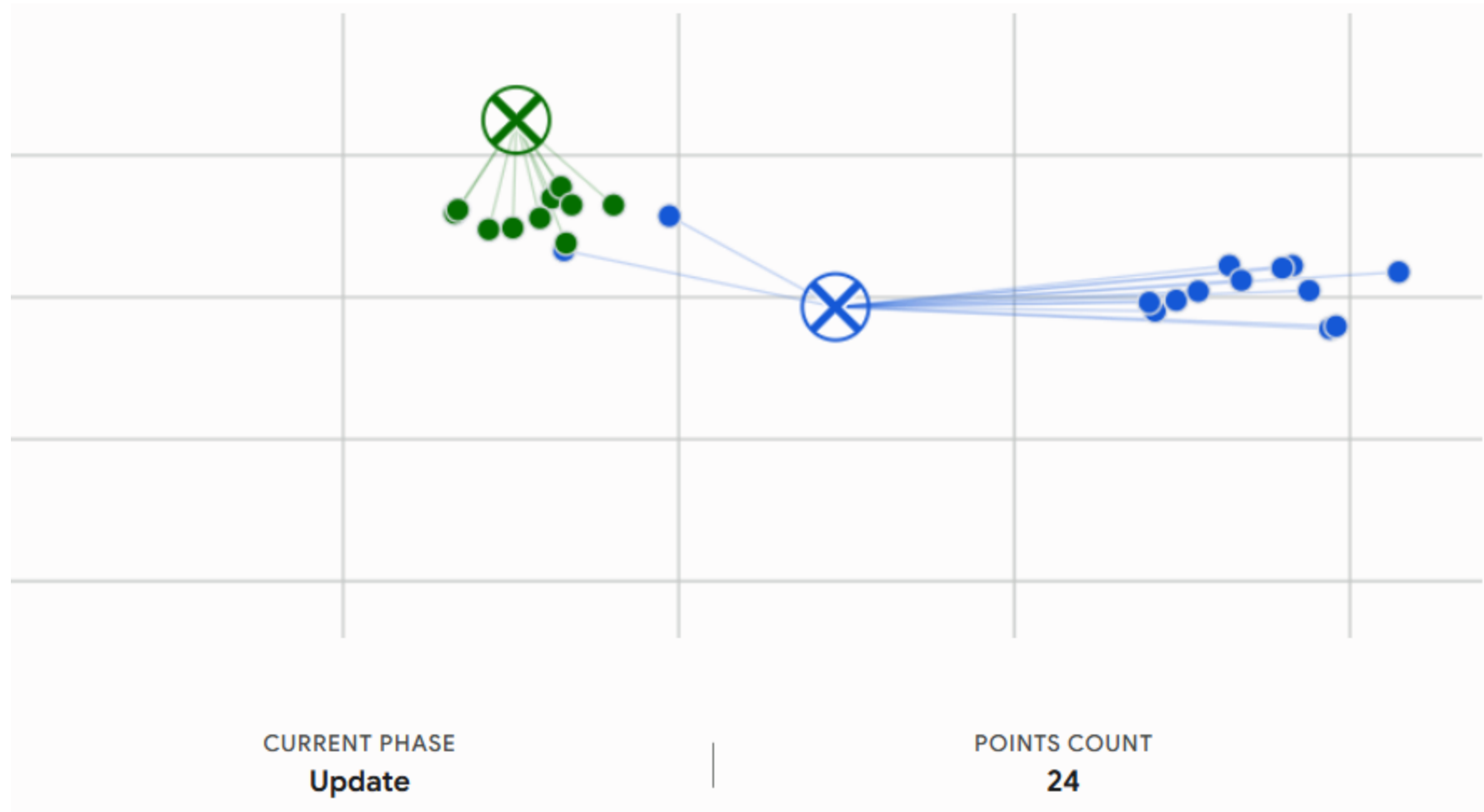
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Visualization 01

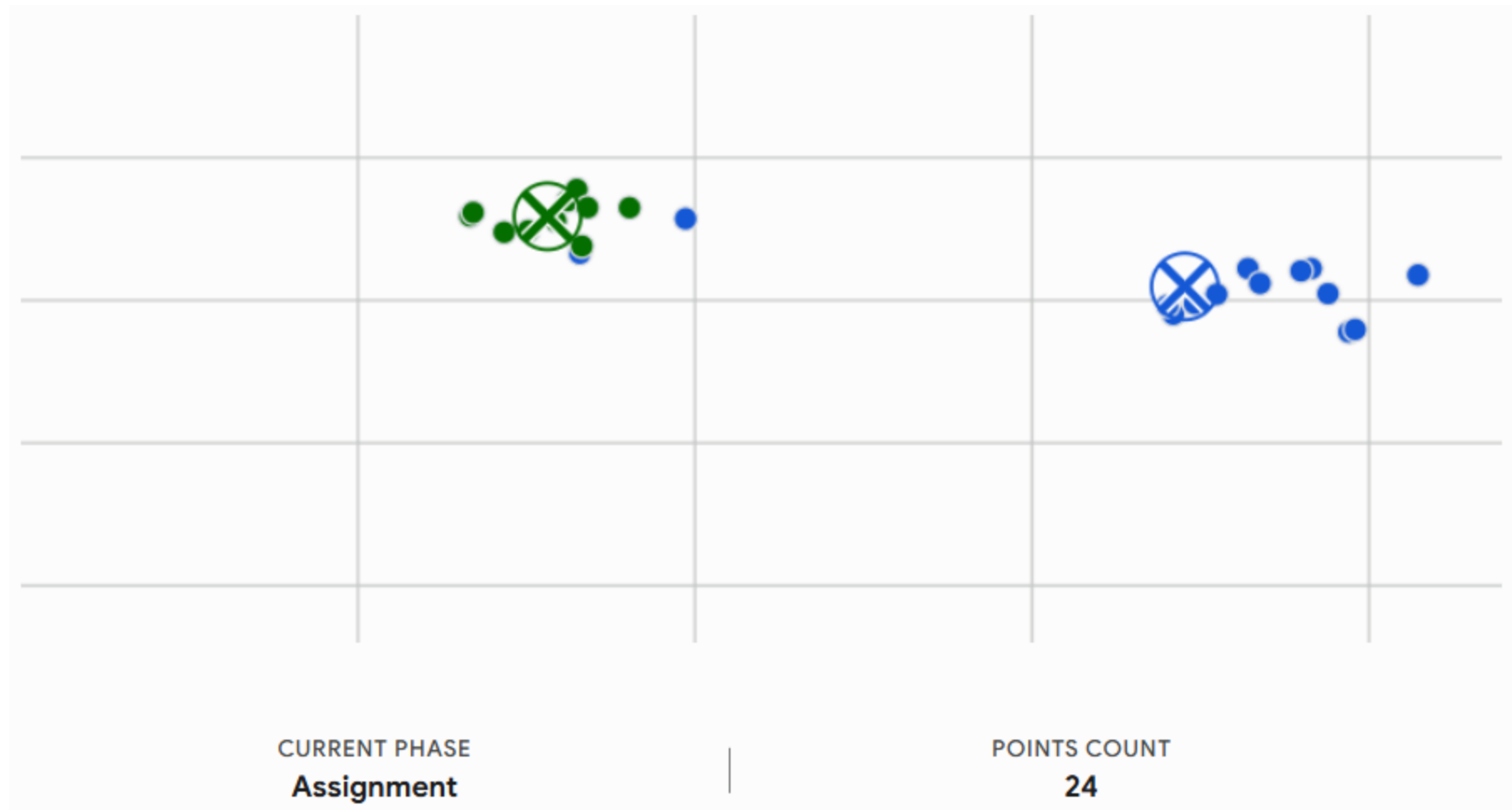
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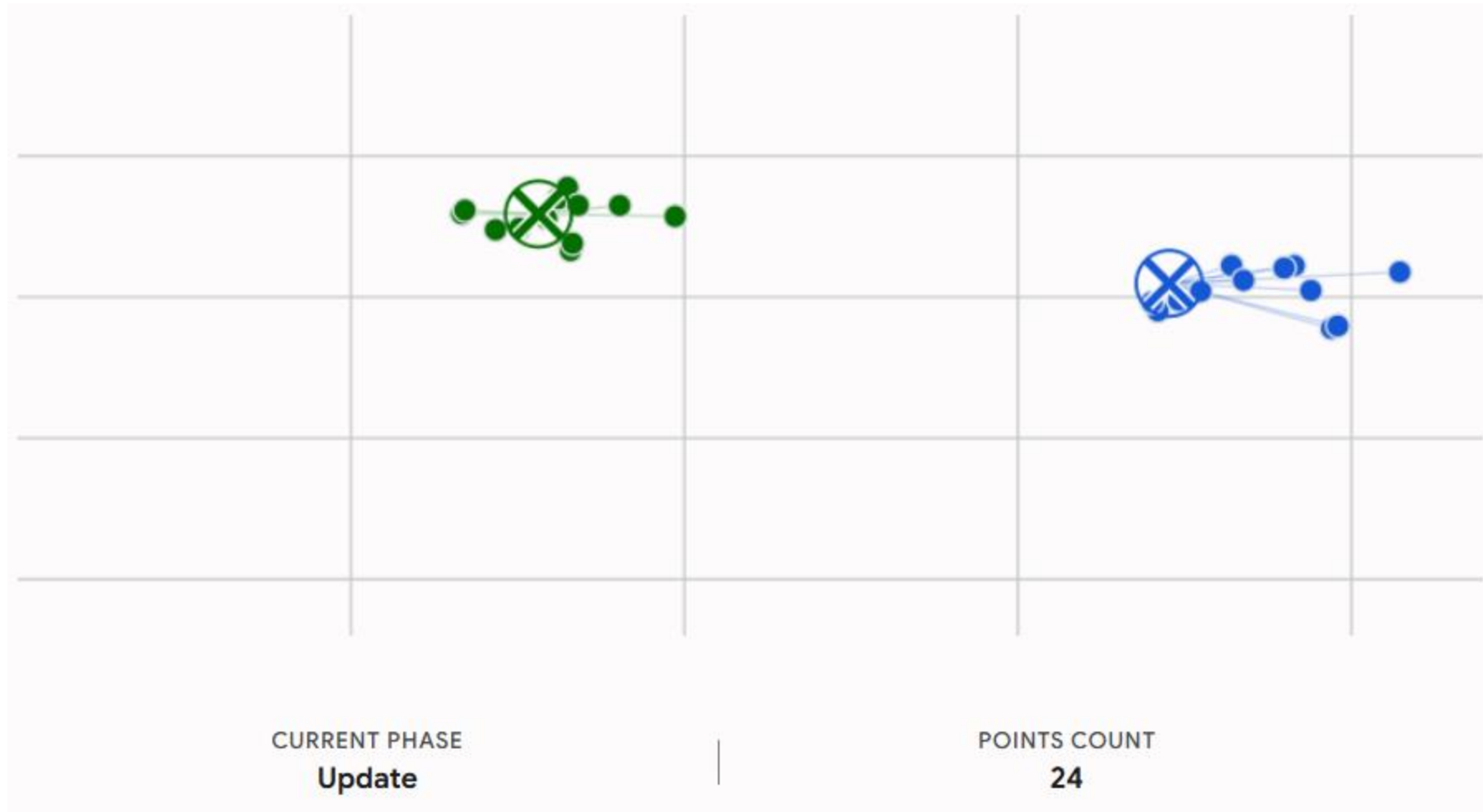
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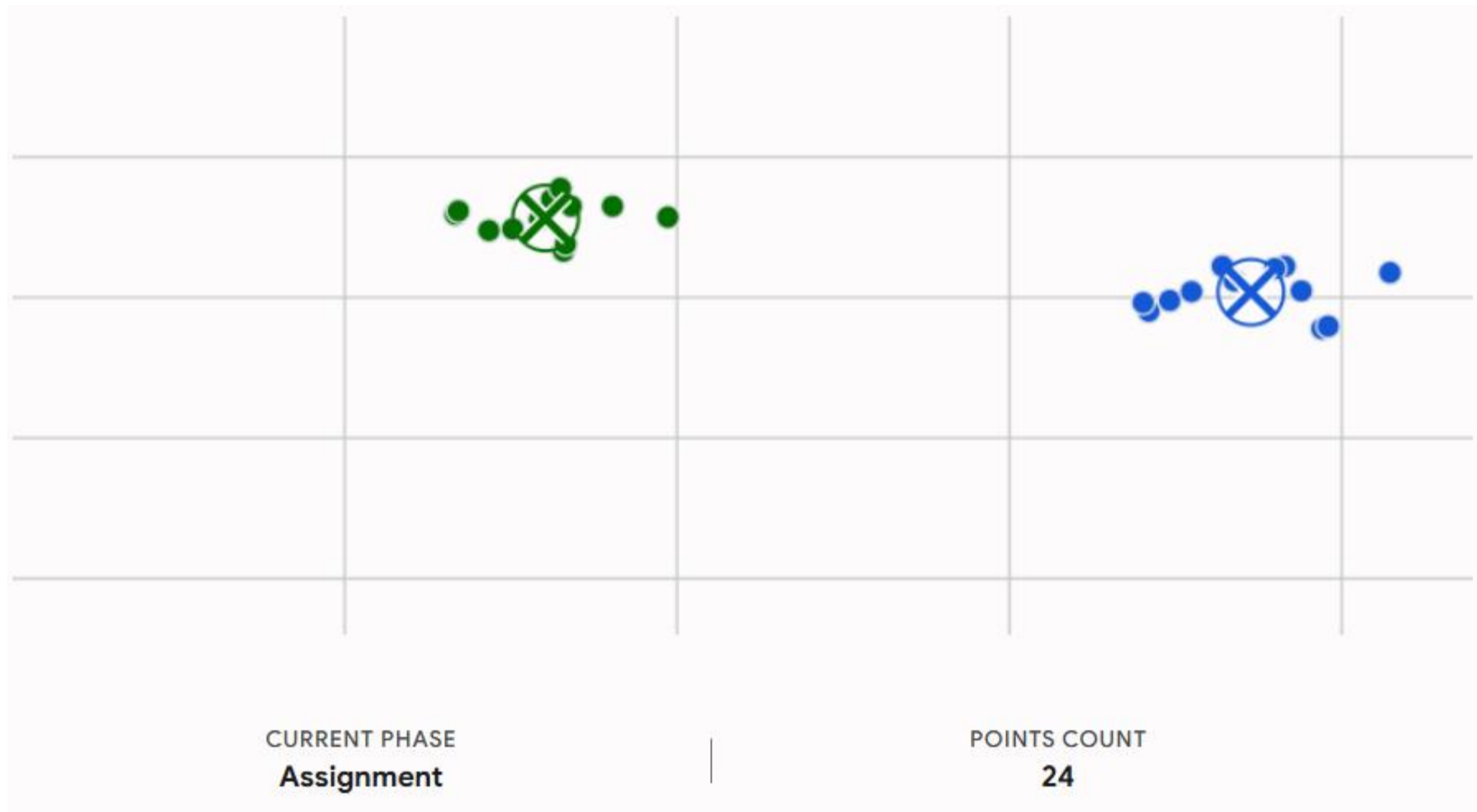
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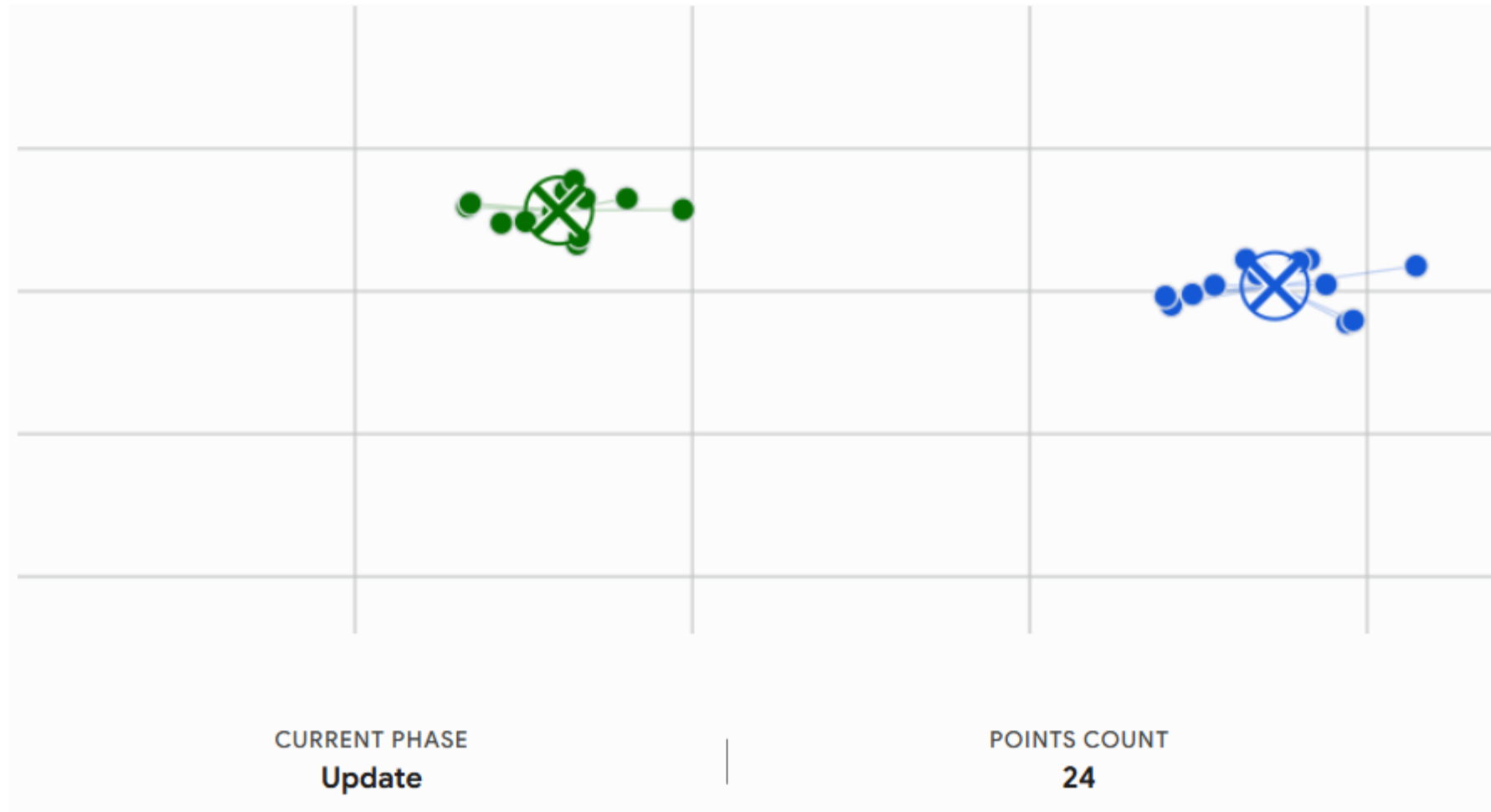
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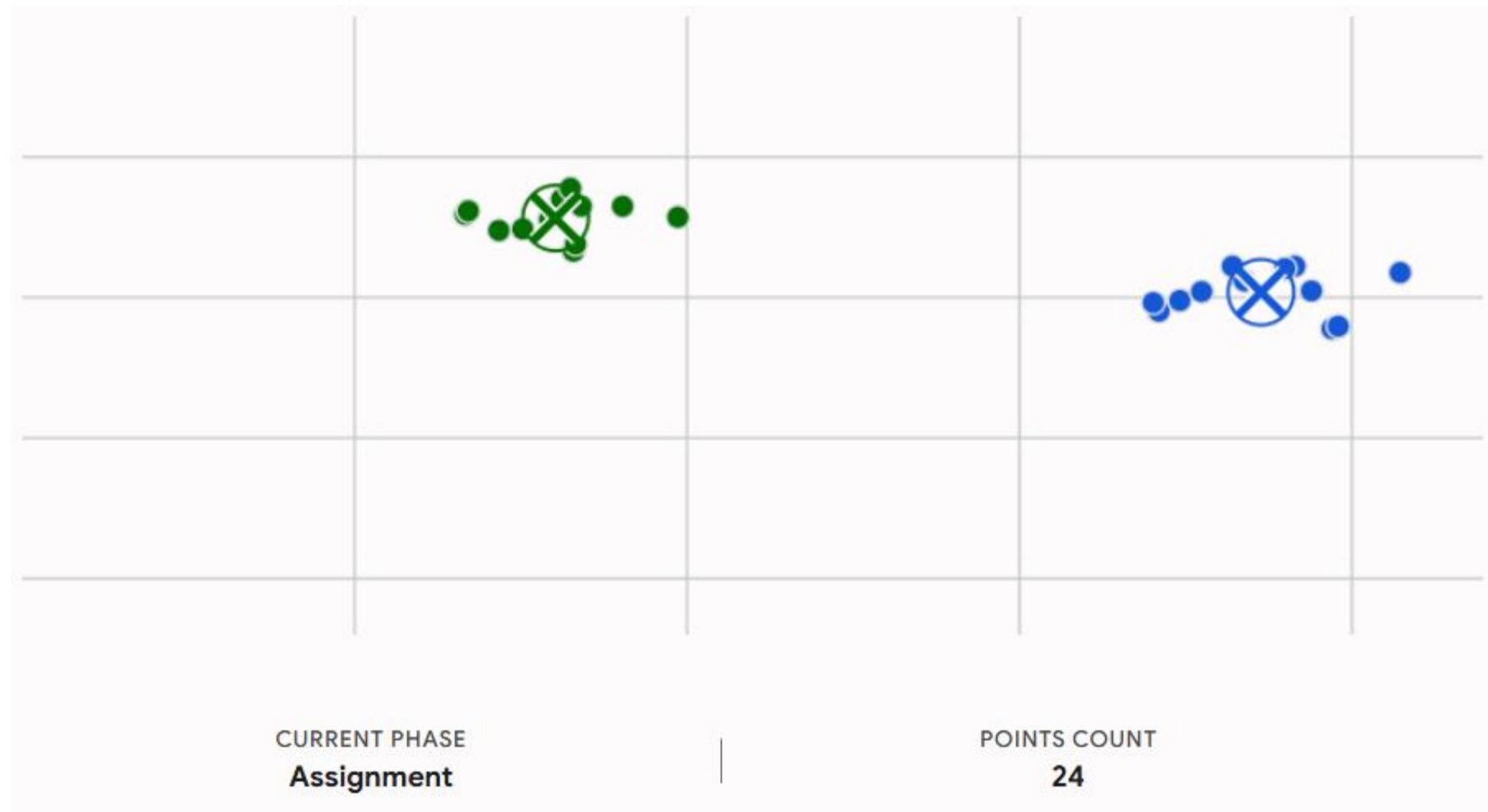
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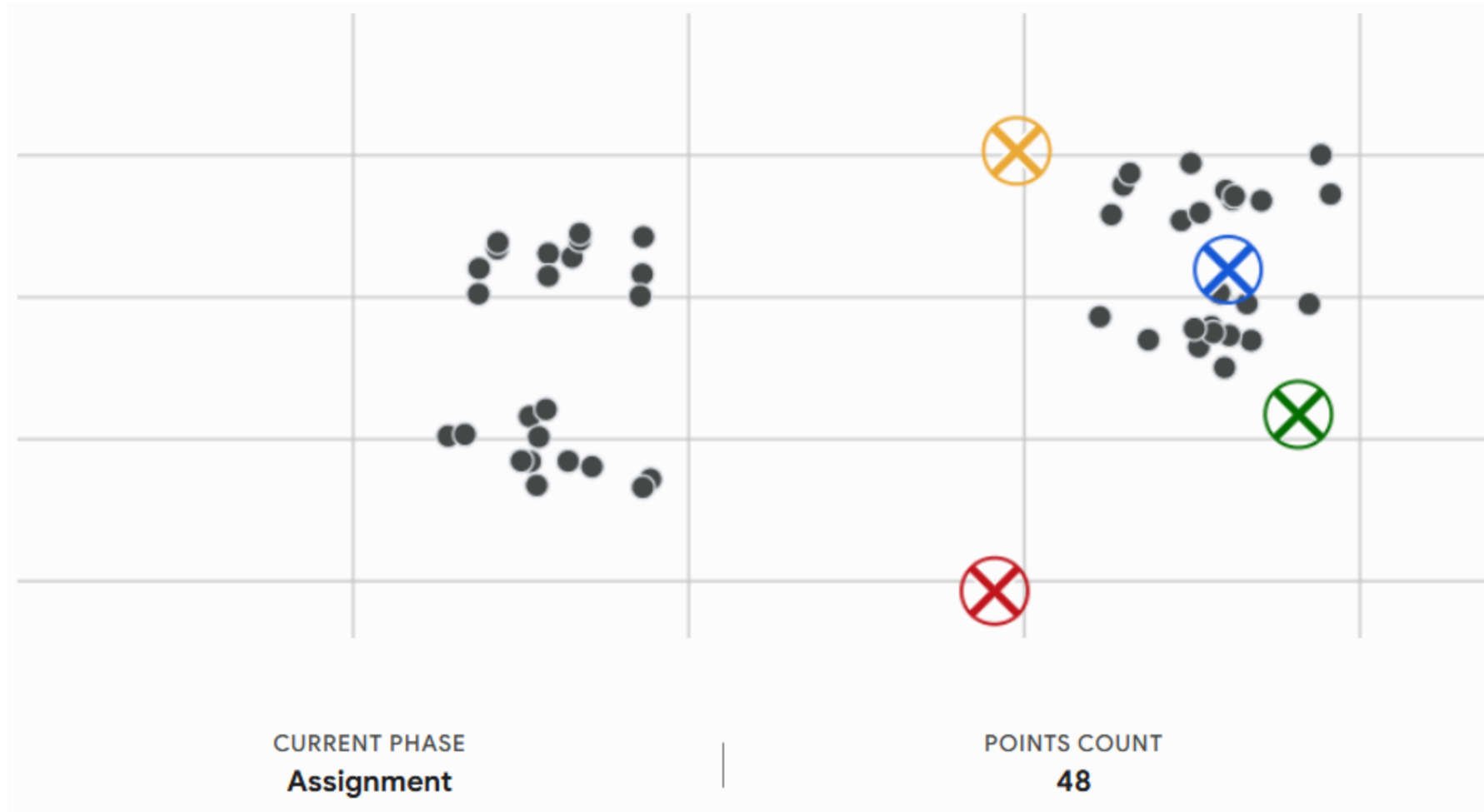
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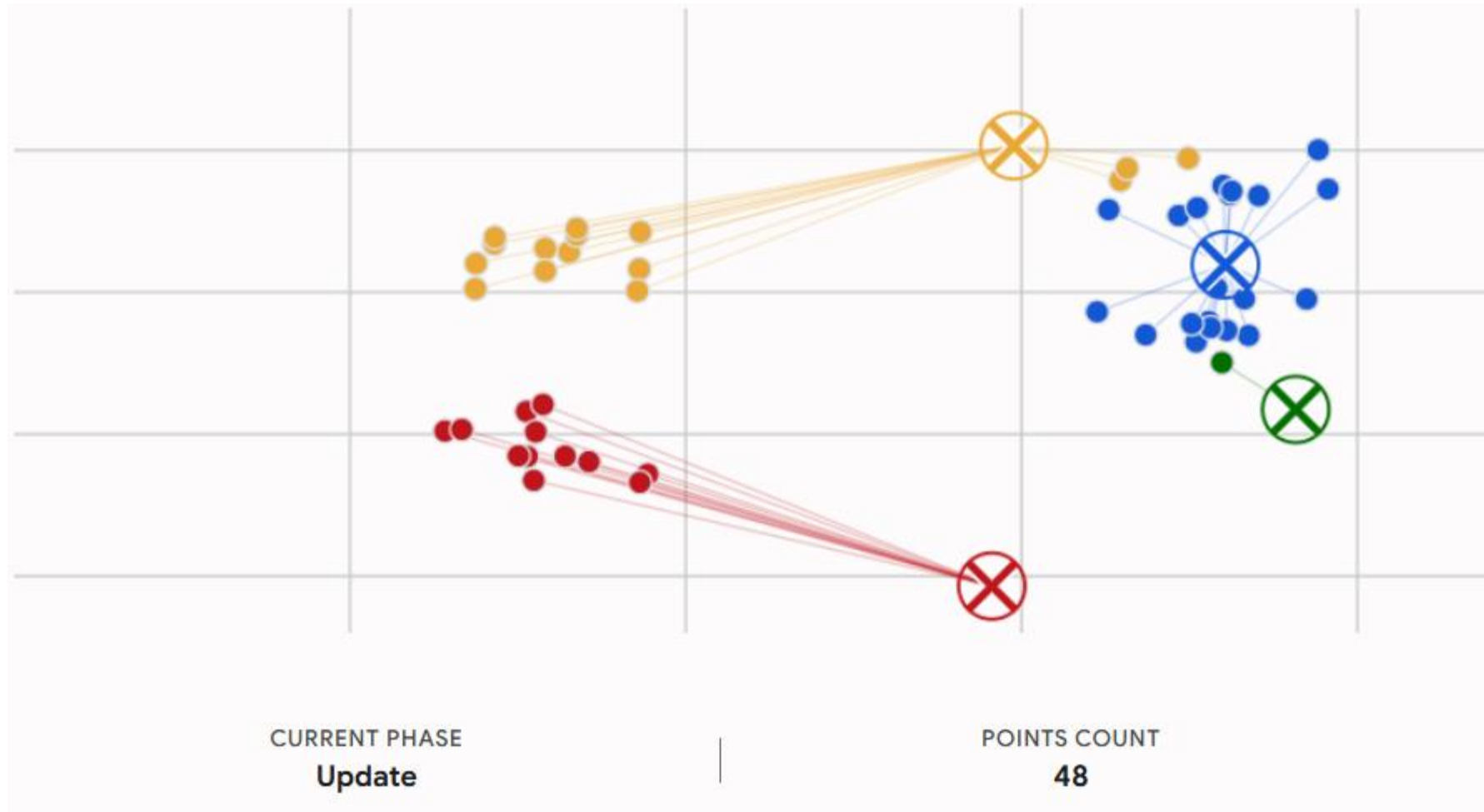
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Visualization 01

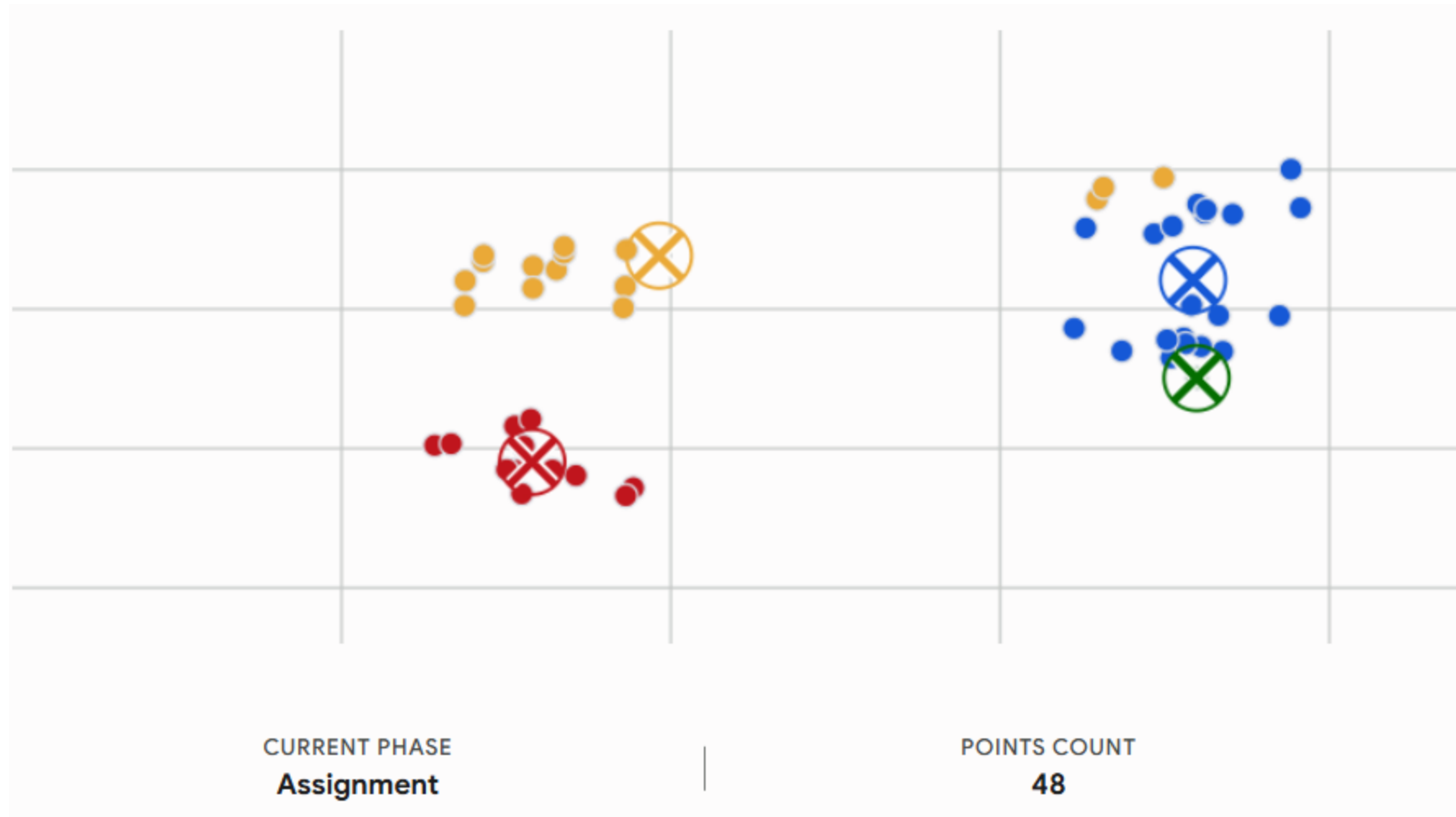
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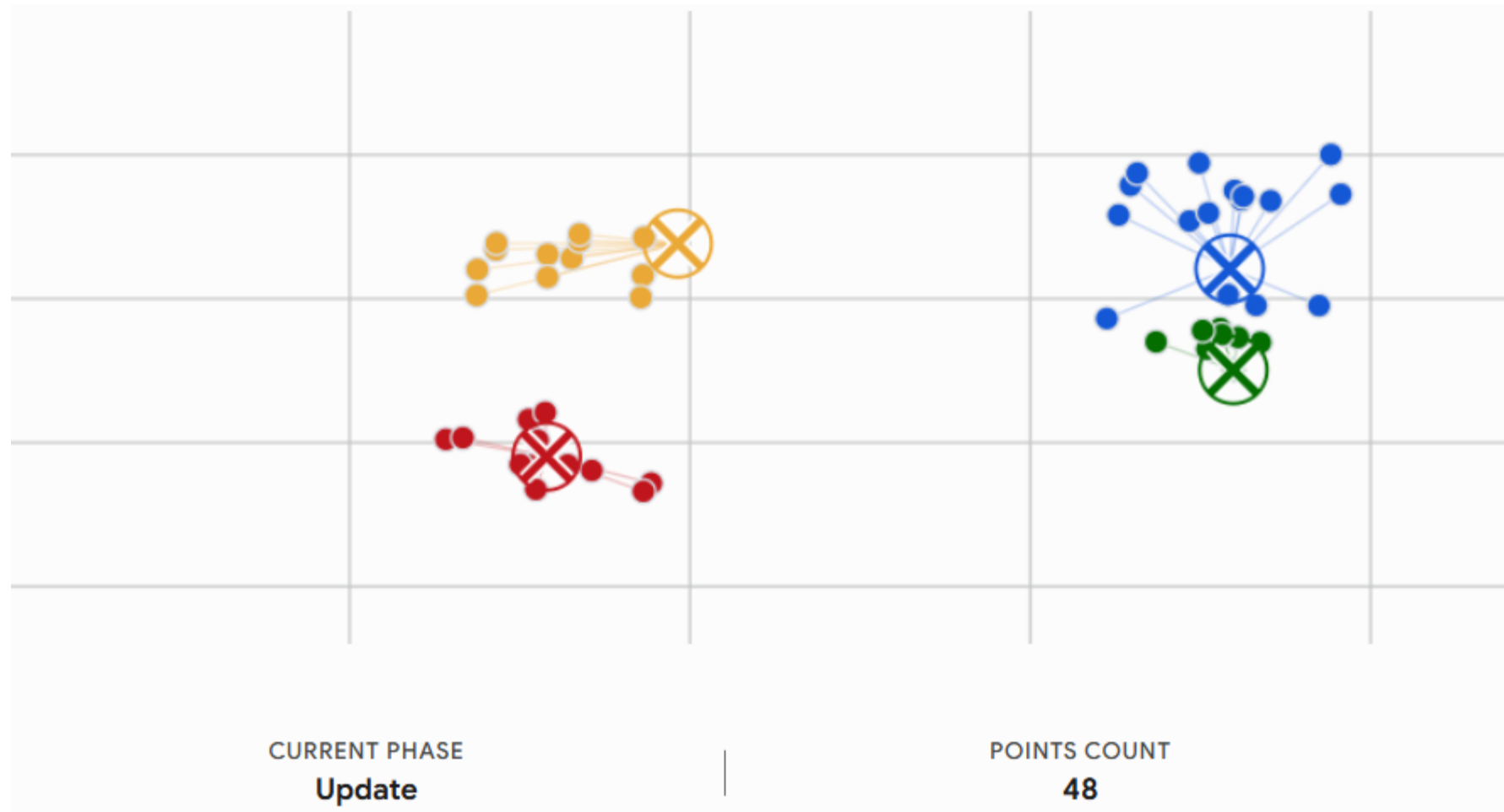
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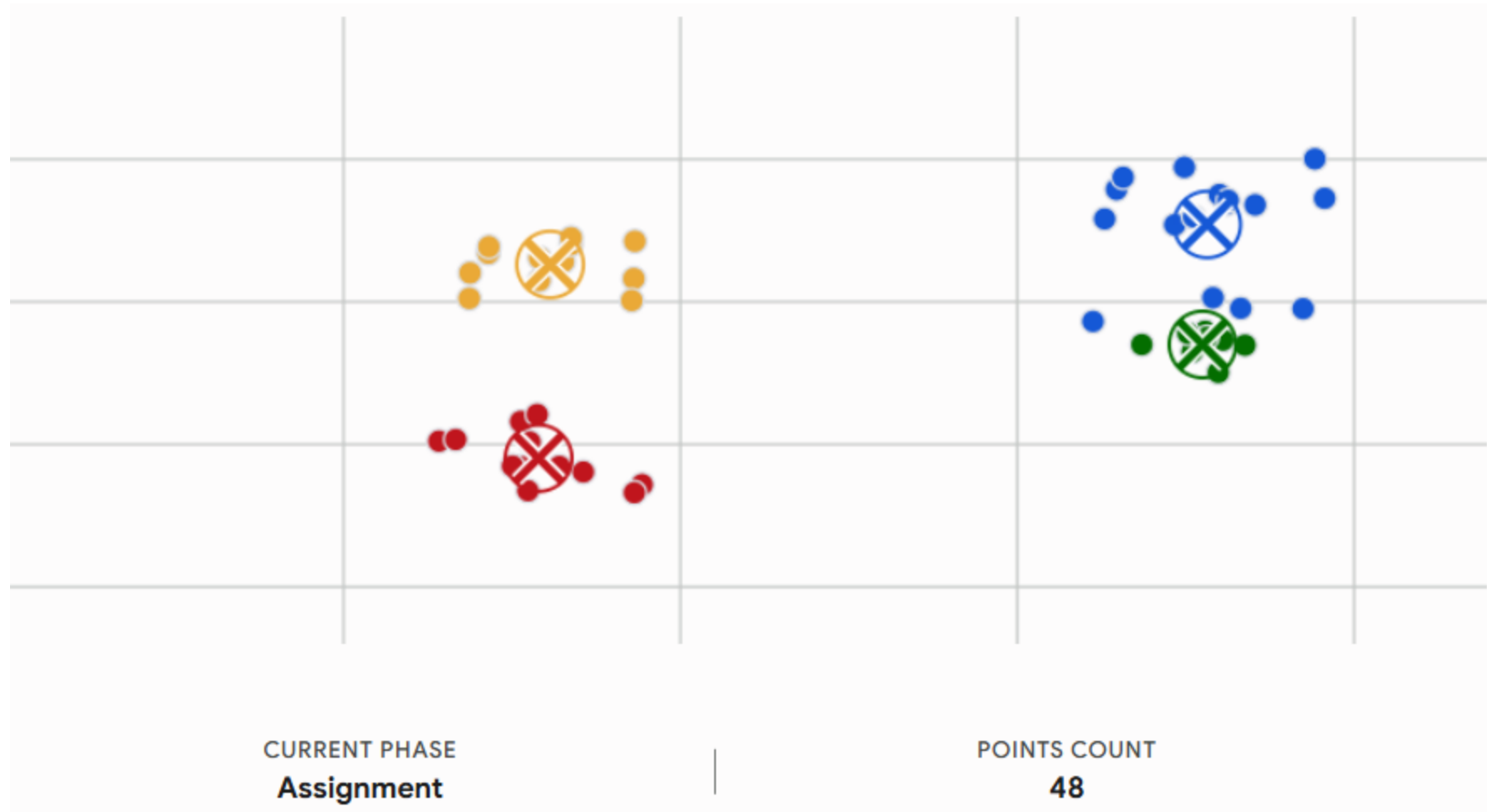
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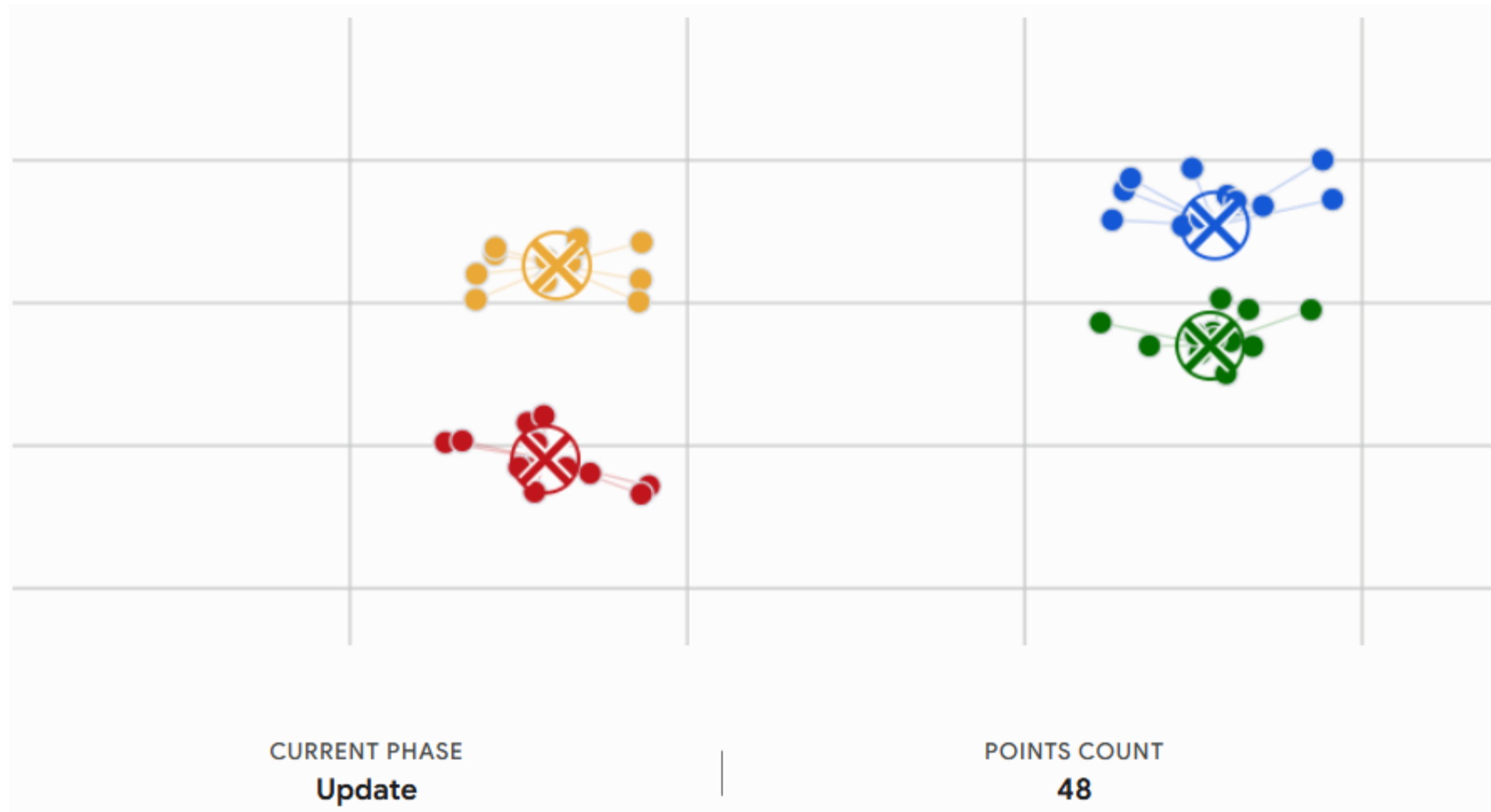
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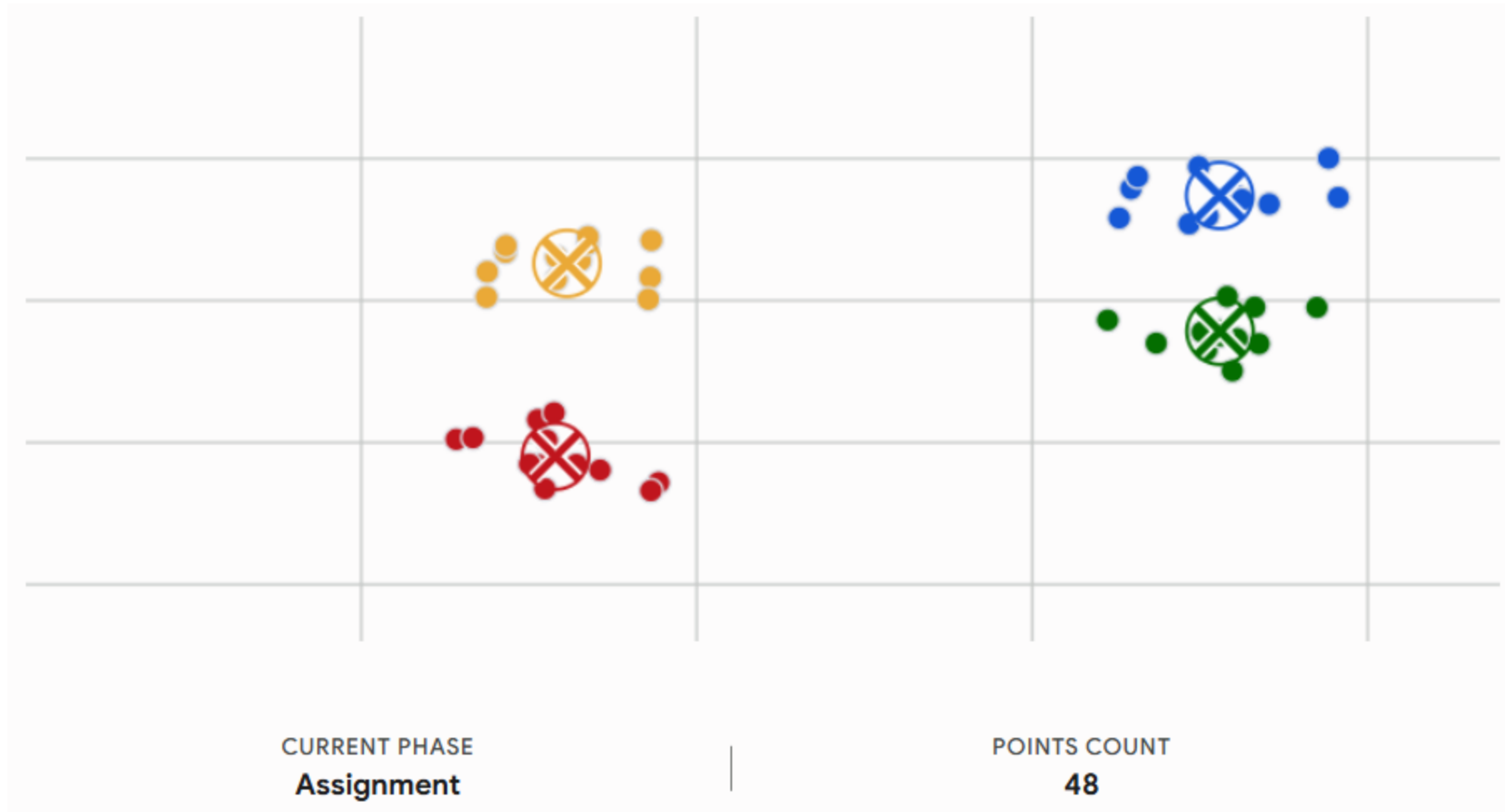
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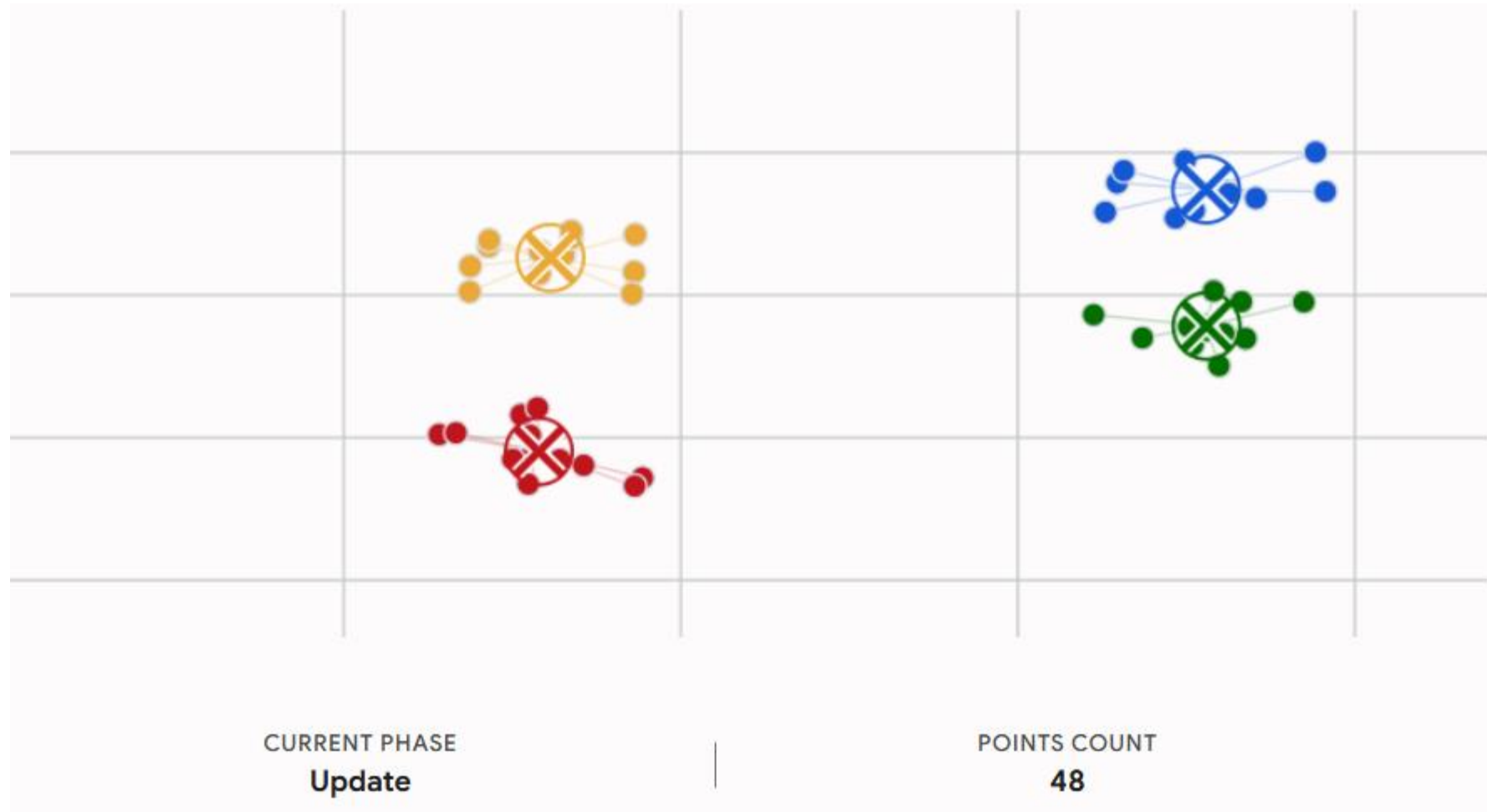
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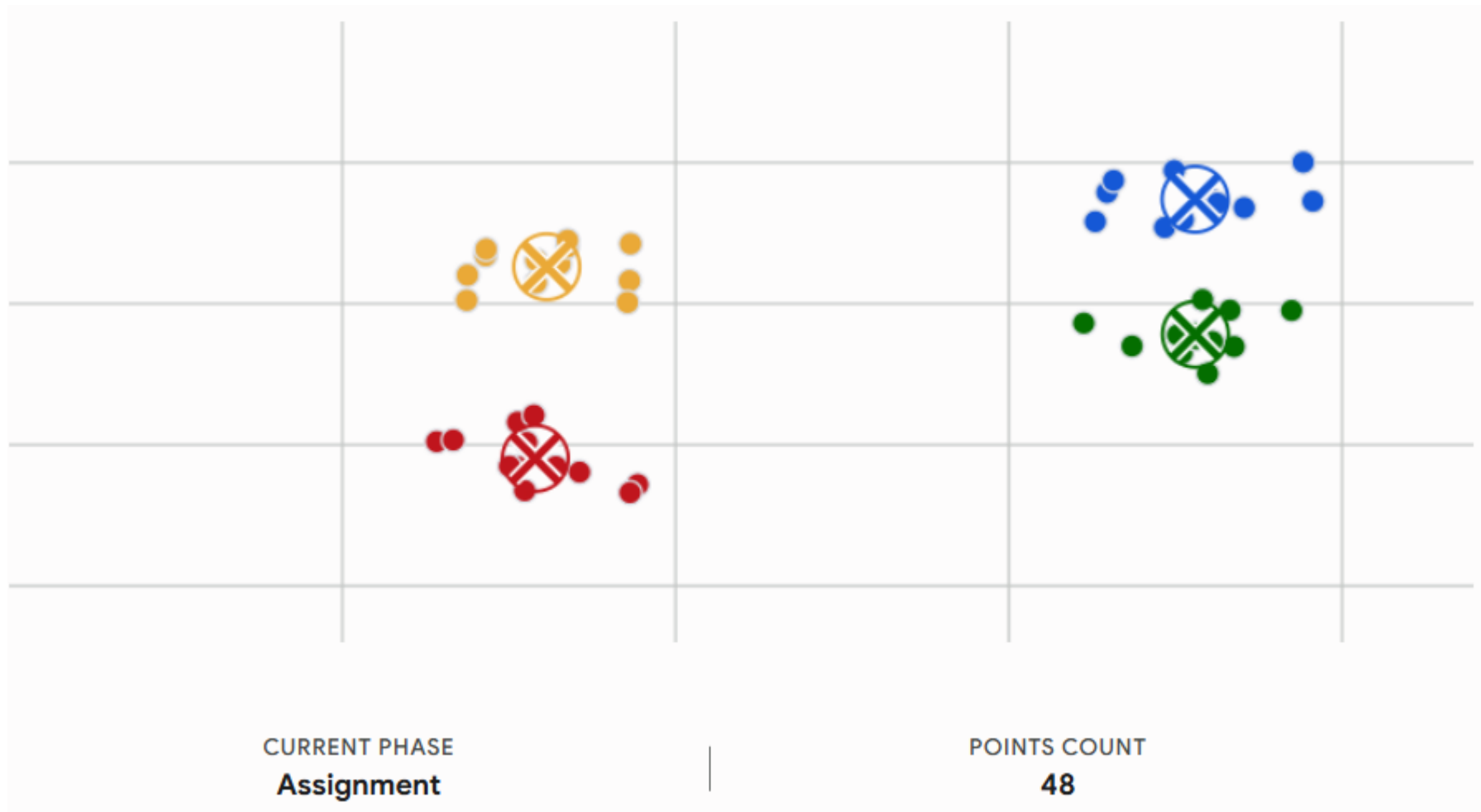
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Thank you very much for listening.